# The CENTRE for EDUCATION in MATHEMATICS and COMPUTING

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## Gauss Contest Grade 7 Solutions

1. The value of $(2+4+6)-(1+3+5)$ is
(A) 0 (B) 3 (C) -3 (D) 21 (E) 111

**Source:** 2017 Gauss Grade 7 #1

**Primary Topics:** Number Sense

**Secondary Topics:** Operations

**Answer:** B

**Solution:**

Evaluating, $(2+4+6)-(1+3+5)=12-9=3$.

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2. $0.8-0.07$ equals
(A) 0.1 (B) 0.71 (C) 0.793 (D) 0.01 (E) 0.73

**Source:** 2005 Gauss Grade 7 #2

**Primary Topics:** Number Sense

**Secondary Topics:** Decimals | Operations

**Answer:** E

**Solution:**

Calculating, $0.8-0.07=0.80-0.07=0.73$.

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3. $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}$ is equal to
(A) 1 (B) $\frac{1}{64}$ (C) $\frac{3}{14}$ (D) $\frac{7}{8}$ (E) $\frac{3}{8}$

**Source:** 2008 Gauss Grade 7 #3

**Primary Topics:** Number Sense

**Secondary Topics:** Fractions/Ratios

**Answer:** D

**Solution:**

Using a common denominator of 8, we have $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}=\frac{4}{8}+\frac{2}{8}+\frac{1}{8}=\frac{7}{8}$.

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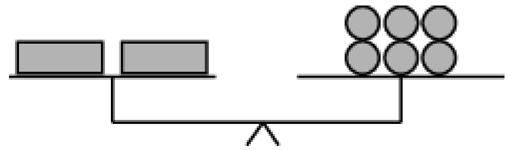
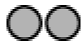
4. The equal-arm scale shown is balanced. One  has the same mass as



(A) 
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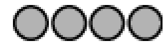
(C) 
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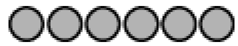
(E) 

**Source:** 2015 Gauss Grade 7 #4

**Primary Topics:** Algebra and Equations

**Secondary Topics:** Equations Solving

**Answer:** B

**Solution:**

The equal-arm balance shows that 2 rectangles have the same mass as 6 circles. If we organize these shapes into two equal piles on both sides of the balance, then we see that 1 rectangle has the same mass as 3 circles.

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5. If there is no tax, which of the following costs more than \$18 to purchase?
(A) Five \$1 items and five \$2 items
(B) Nine \$1 items and four \$2 items
(C) Nine \$1 items and five \$2 items
(D) Two \$1 items and six \$2 items
(E) Sixteen \$1 items and no \$2 items

**Source:** 2018 Gauss Grade 7 #5

**Primary Topics:** Number Sense

**Secondary Topics:** Operations

**Answer:** C

**Solution:**

The cost of nine \$1 items and five \$2 items is $(9 \times \$1) + (5 \times \$2)$, which is $\$9 + \$10$ or \$19.
The correct answer is (C).
(We may check that each of the remaining four answers gives a cost that is less than \$18.)

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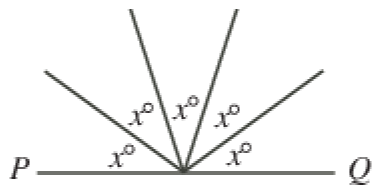
6. If $PQ$ is a straight line, then the value of $x$ is




(A) 36 (B) 72 (C) 18 (D) 20 (E) 45

**Source:** 2008 Gauss Grade 7 #6

**Primary Topics:** Geometry and Measurement

**Secondary Topics:** Angles

**Answer:** A

**Solution:**

Since $PQ$ is a straight line, then $x^\circ + x^\circ + x^\circ + x^\circ + x^\circ = 180^\circ$ or $5x = 180$ or $x = 36$.

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7. $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3}$ equals

(A) $3\frac{1}{3}$ (B) $7 + \frac{1}{3}$ (C) $\frac{3}{7}$ (D) $7 + 3$ (E) $7 \times \frac{1}{3}$

**Source:** 2011 Gauss Grade 7 #7

**Primary Topics:** Number Sense

**Secondary Topics:** Fractions/Ratios

**Answer:** E

**Solution:**

Since we are adding $\frac{1}{3}$ seven times, then the result is equal to $7 \times \frac{1}{3}$.

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8. A circle has a radius of 4 cm. A line segment joins two points on the circle. What is the greatest possible length of the line segment?
   (A) 10 cm (B) 8 cm (C) 4 cm (D) 12 cm (E) 6 cm

**Source:** 2023 Gauss Grade 7 #8

**Primary Topics:** Geometry and Measurement

**Secondary Topics:** Circles | Measurement

**Answer:** B

**Solution:**

The line segment with greatest length that joins two points on a circle is a diameter of the circle. Since the circle has a radius of 4 cm, then its diameter has length $2 \times 4\text{ cm} = 8$ cm, and so the greatest possible length of the line segment is 8 cm.

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9. The word BANK is painted exactly as shown on the outside of a clear glass window. Looking out through the window from the inside of the building, the word appears as
   (A) BAИꓘ (B) KNAᗺ (C) ᗺAИꓘ (D) ꓘИAᗺ (E) KNAB

**Source:** 2007 Gauss Grade 7 #9

**Primary Topics:** Geometry and Measurement

**Secondary Topics:** Transformations

**Answer:** D

**Solution:**

When the word BANK is viewed from the inside of the window, the letters appear in the reverse order and the letters themselves are all backwards, so the word appears as BANK.

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10. Which one of the following is equal to 17?

(A) $3 - 4 \times 5 + 6$ (B) $3 \times 4 + 5 \div 6$ (C) $3 + 4 \times 5 - 6$
(D) $3 \div 4 + 5 - 6$ (E) $3 \times 4 \div 5 + 6$

**Source:** 2014 Gauss Grade 7 #10

**Primary Topics:** Number Sense

**Secondary Topics:** Operations

**Answer:** C

**Solution:**

In the table below, each of the five expressions is evaluated using the correct order of operations.

| Expression | Value |
|---|---|
| $(A)\ 3 - 4 \times 5 + 6$ | $3 - 20 + 6 = -17 + 6 = -11$ |
| $(B)\ 3 \times 4 + 5 \div 6$ | $12 + 5 \div 6 = 12 + \frac{5}{6} = 12\frac{5}{6}$ |
| $(C)\ 3 + 4 \times 5 - 6$ | $3 + 20 - 6 = 23 - 6 = 17$ |
| $(D)\ 3 \div 4 + 5 - 6$ | $\frac{3}{4} + 5 - 6 = 5\frac{3}{4} - 6 = \frac{23}{4} - \frac{24}{4} = -\frac{1}{4}$ |
| $(E)\ 3 \times 4 \div 5 + 6$ | $12 \div 5 + 6 = \frac{12}{5} + 6 = 2\frac{2}{5} + 6 = 8\frac{2}{5}$ |

The only expression that is equal to 17 is $3 + 4 \times 5 - 6$, or (C).

11. The sum of the prime factors of 42 is

(A) 23 (B) 43 (C) 12 (D) 17 (E) 13

**Source:** 2022 Gauss Grade 7 #11

**Primary Topics:** Number Sense

**Secondary Topics:** Prime Numbers | Divisibility | Factoring

**Answer:** C

**Solution:**

Since 42 is an even number, then 2 is a factor of 42.
Since $42 = 2 \times 21$ and $21 = 3 \times 7$, then $42 = 2 \times 3 \times 7$.
Each of $2, 3$ and 7 is a prime number and each is a factor of 42.
Thus, the sum of the prime factors of 42 is $2 + 3 + 7 = 12$.
(We note that $1, 6, 14, 21,$ and 42 are also factors of 42, however they are not prime numbers.)

12. A bamboo plant grows at a rate of 105 cm per day. On May 1st at noon it was 2 m tall. Approximately how tall, in metres, was it on May 8th at noon?
(A) 10.4 (B) 8.3 (C) 3.05 (D) 7.35 (E) 9.35

**Source:** 2005 Gauss Grade 7 #12

**Primary Topics:** Number Sense

**Secondary Topics:** Rates | Decimals | Estimation | Measurement

**Answer:** E

**Solution:**

Since the bamboo plant grows at a rate of 105 cm per day and there are 7 days from May 1st and May 8th, then it grows $7 \times 105 = 735$ cm in this time period.
Since $735 \text{ cm} = 7.35 \text{ m}$, then the height of the plant on May 8th is $2 + 7.35 = 9.35$ m.

13. If the mean (average) of five consecutive integers is 21, the smallest of the five integers is

(A) 17 (B) 21 (C) 1 (D) 18 (E) 19

**Source:** 2010 Gauss Grade 7 #13

**Primary Topics:** Number Sense

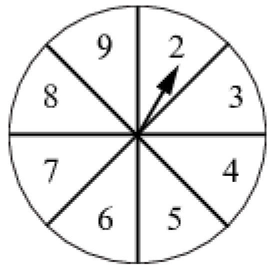
**Secondary Topics:** Patterning/Sequences/Series | Averages

**Answer:** E

**Solution:**

Solution 1
The mean of 5 consecutive integers is equal to the number in the middle.
Since the numbers have a mean of 21, if we were to distribute the quantities equally, we would have 21, 21, 21, 21, and 21.
Since the numbers are consecutive, the second number is 1 less than the 21 in the middle, while the fourth number is 1 more than the 21 in the middle.
Similarly, the first number is 2 less than the 21 in the middle, while the fifth number is 2 more than the 21 in the middle.
Thus, the numbers are $21-2$, $21-1$, $21$, $21+1$, $21+2$.
The smallest of 5 consecutive integers having a mean of 21, is 19.
Solution 2
Since 21 is the mean of five consecutive integers, the smallest of these five integers must be less than 21.
Suppose that the smallest number is 20.
The mean of $20, 21, 22, 23$, and $24$ is $\dfrac{20+21+22+23+24}{5}=22$.
This mean of 22 is greater than the required mean of 21; thus, the smallest of the 5 consecutive integers must be less than 20.
Suppose that the smallest number is 19.
The mean of $19, 20, 21, 22$, and $23$, is $\dfrac{19+20+21+22+23}{5}=21$, as required.
Thus, the smallest of the 5 consecutive integers is 19.

14. A circular spinner is divided into 8 equal sections, as shown. An arrow is attached to the centre of the spinner. The arrow is spun once. What is the probability that the arrow stops in a section containing a prime number that is odd?




(A) $\frac{1}{8}$ (B) $\frac{2}{8}$ (C) $\frac{3}{8}$ (D) $\frac{4}{8}$ (E) $\frac{7}{8}$

**Source:** 2019 Gauss Grade 7 #14

**Primary Topics:** Counting and Probability

**Secondary Topics:** Probability | Prime Numbers

**Answer:** C

**Solution:**

On the spinner given, the prime numbers that are odd are $3, 5$ and $7$.
Since the spinner is divided into 8 equal sections, the probability that the arrow stops in a section containing a prime number that is odd is $\frac{3}{8}$.

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15. Suppose $n$ is a number such that the mean (average) of the list of numbers $2, 9, 4, n, 2n$ is equal to $6$. What is the value of $n$?
(A) 9 (B) 12 (C) 10 (D) 5 (E) 6

**Source:** 2023 Gauss Grade 7 #15

**Primary Topics:** Number Sense | Data Analysis

**Secondary Topics:** Averages | Expressions

**Answer:** D

**Solution:**

The given list, $2, 9, 4, n, 2n$ contains 5 numbers. The average of these 5 numbers is 6, and so the sum of the 5 numbers is $5 \times 6 = 30$. That is, $2 + 9 + 4 + n + 2n = 30$ or $15 + 3n = 30$, and so $3n = 15$ or $n = 5$.

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16. A standard fair coin is tossed three times. What is the probability that the three outcomes are all the same?
(A) $\frac{1}{2}$ (B) $\frac{3}{16}$ (C) $\frac{1}{4}$ (D) $\frac{5}{16}$ (E) $\frac{1}{8}$

**Source:** 2022 Gauss Grade 7 #16

**Primary Topics:** Counting and Probability

**Secondary Topics:** Probability | Counting | Fractions/Ratios

**Answer:** C

**Solution:**

*Solution 1*:

When a standard fair coin is tossed three times, there are 8 possible outcomes.
If H represents head and T represents tail, these 8 outcomes are: HHH, HHT, HTH, THH, HTT, THT, TTH, and TTT.
Of these 8 outcomes, there are exactly 2 whose outcomes are all the same (HHH and TTT).
Therefore, the probability that the three outcomes are all the same is $\frac{2}{8} = \frac{1}{4}$.

*Solution 2*:

The three outcomes are all the same exactly when each of the three tosses is a head or when each of the three tosses is a tail.
When a coin is tossed, the probability that the coin shows a head is $\frac{1}{2}$ and the probability that the coin shows a tail is also $\frac{1}{2}$ (there are two possible outcomes and each is equally probable).
Thus, the probability that each of the three tosses is a head is $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$.
Similarly, the probability that each of the three tosses is a tail is $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$.
Therefore, the probability that the three outcomes are all the same is $\frac{1}{8} + \frac{1}{8} = \frac{2}{8} = \frac{1}{4}$.

17. The ratio of junior kindergarteners to senior kindergarteners at Gauss Public School is 8 : 5. If there are 128 junior kindergarteners at the school, then how many kindergarteners are there at the school?
(A) 218 (B) 253 (C) 208 (D) 133 (E) 198

**Source:** 2012 Gauss Grade 7 #17

**Primary Topics:** Number Sense

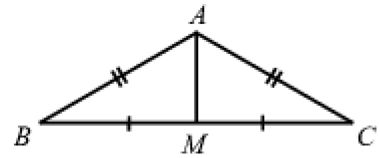
**Secondary Topics:** Fractions/Ratios

**Answer:** C

**Solution:**

Solution 1
Since the ratio of JKs to SKs is 8 : 5, then for every 5 SKs there are 8 JKs.
That is, the number of SKs at Gauss Public School is $\frac{5}{8}$ of the number of JKs.
Since the number of JKs at the school is 128, the number of SKs is $\frac{5}{8} \times 128 = \frac{640}{8} = 80$.
The number of kindergarteners at the school is the number of JKs added to the number of SKs or $128 + 80 = 208$.
Solution 2
Since the ratio of JKs to SKs is 8 : 5, then for every 8 JKs there are $8 + 5 = 13$ students.
That is, the number of kindergarteners at Gauss Public School is $\frac{13}{8}$ of the number of JKs.
Since the number of JKs at the school is 128, the number of kindergarteners is
$\frac{13}{8} \times 128 = \frac{1664}{8} = 208$.

18. In the diagram, $\triangle ABC$ is isosceles. $M$ is on $BC$ so that $BM = MC$. If the perimeter of $\triangle ABC$ is 64 and the perimeter of $\triangle ABM$ is 40, what is the length of $AM$?




(A) 10 (B) 8 (C) 16 (D) 12 (E) 24

**Source:** 2021 Gauss Grade 7 #18

**Primary Topics:** Geometry and Measurement

**Secondary Topics:** Measurement | Triangles

**Answer:** B

**Solution:**

The perimeter of $\triangle ABC$ is equal to $AB + BC + CA$ or $AB + BM + MC + CA$.
Since $AB = CA$ and $BM = MC$, then half of the perimeter of $\triangle ABC$ is equal to $AB + BM$.
The perimeter of $\triangle ABC$ is 64 and so $AB + BM = \frac{1}{2} \times 64 = 32$.
The perimeter of $\triangle ABM$ is 40, and so $AM + AB + BM = 40$.
Since $AB + BM = 32$, then $AM = 40 - 32 = 8$.

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19. Which of the following is closest to one million ($10^6$) seconds?
(A) 1 day (B) 10 days (C) 100 days (D) 1 year (E) 10 years

**Source:** 2006 Gauss Grade 7 #19

**Primary Topics:** Number Sense

**Secondary Topics:** Measurement | Estimation

**Answer:** B

**Solution:**

Solution 1
In one minute, there are 60 seconds.
In one hour, there are 60 minutes, so there are $60 \times 60 = 3600$ seconds.
In one day, there are 24 hours, so there are $24 \times 3600 = 86\,400$ seconds.
Therefore, $10^6$ seconds is equal to $\frac{10^6}{86\,400} \approx 11.574$ days, which of the given choices is closest to 10 days.

Solution 2
Since there are 60 seconds in one minute, then $10^6$ seconds is $\frac{10^6}{60} \approx 16\,666.67$ minutes.
Since there are 60 minutes in one hour, then $16\,666.67$ minutes is $\frac{16\,666.67}{60} \approx 277.78$ hours.
Since there are 24 hours in one day, then 277.78 hours is $\frac{277.78}{24} \approx 11.574$ days, which of the given choices is closest to 10 days.

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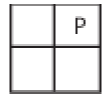
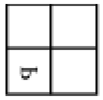
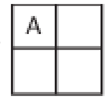
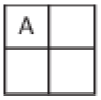
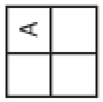
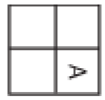
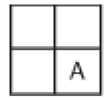
20. The letter P is written in a $2 \times 2$ grid of squares as shown: 

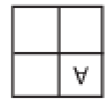
A combination of rotations about the centre of the grid and reflections in the two lines through the centre achieves the result: 

When the same combination of rotations and reflect is applied to 

, the result is

(A) 

(B) 
(C) 
(D) 

(E) 

**Source:** 2006 Gauss Grade 7 #20

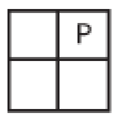
**Primary Topics:** Geometry and Measurement

**Secondary Topics:** Transformations

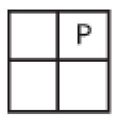
**Answer:** B

**Solution:**

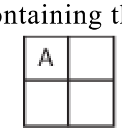
One possible way to transform the initial position of the P to the final position of the P is to reflect the grid in the vertical line in the middle to obtain



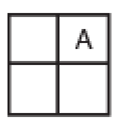

and then rotate the grid 90° counterclockwise about the centre to obtain




Applying these transformations to the grid containing the A, we obtain




and then




(There are many other possible combinations of transformations which will produce the same resulting image with the P; each of these combinations will produce the same result with the A.)

21. Kathy owns more cats than Alice and more dogs than Bruce. Alice owns more dogs than Kathy and fewer cats than Bruce. Which of the statements *it must* be true?
    (A) Bruce owns the fewest cats.
    (B) Bruce owns the most cats.
    (C) Kathy owns the most cats.
    (D) Alice owns the most dogs.
    (E) Kathy owns the fewest dogs.

**Source:** 2019 Gauss Grade 7 #21

**Primary Topics:** Other | Algebra and Equations

**Secondary Topics:** Logic | Inequalities

**Answer:** D

**Solution:**

We begin by separating the given information, as follows:

- Kathy owns more cats than Alice
- Kathy owns more dogs than Bruce
- Alice owns more dogs than Kathy
- Bruce owns more cats than Alice

From bullets 2 and 3, we can conclude that Alice owns more dogs than both Kathy and Bruce.
From bullet 4, we can conclude that answer (A) is not true.
From bullets 1 and 4, we can conclude that both Kathy and Bruce own more cats than Alice.
However, we cannot determine if Kathy owns more cats than Bruce, or vice versa.
Therefore, we cannot conclude that (B) or (C) *must* be true.
From bullet 2, we can conclude that (E) is not true.
Thus the statement which *must* be true is (D).

Kathy owns more cats than Alice
Kathy owns more dogs than Bruce
Alice owns more dogs than Kathy
Bruce owns more cats than Alice

22. In a bin at the Gauss Grocery, the ratio of the number of apples to the number of oranges is 1 : 4, and the ratio of the number of oranges to the number of lemons is 5 : 2. What is the ratio of the number of apples to the number of lemons?
(A) 1 : 2 (B) 4 : 5 (C) 5 : 8 (D) 20 : 8 (E) 2 : 1

**Source:** 2005 Gauss Grade 7 #22

**Primary Topics:** Number Sense

**Secondary Topics:** Fractions/Ratios

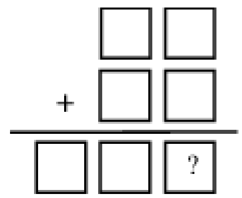
**Answer:** C

**Solution:**

Solution 1
We start by assuming that there are 20 oranges. (We pick 20 since the ratio of apples to oranges is 1 : 4 and the ratio of oranges to lemons is 5 : 2, so we pick a number of oranges which is divisible by 4 and by 5. Note that we did not have to assume that there were 20 oranges, but making this assumption makes the calculations much easier.)
Since there are 20 oranges and the ratio of the number of apples to the number of oranges is 1 : 4, then there are $\frac{1}{4} \times 20 = 5$ apples.
Since there are 20 oranges and the ratio of the number of oranges to the number of lemons is 5 : 2, then there are $\frac{2}{5} \times 20 = 8$ lemons.
Therefore, the ratio of the number of apples to the number of lemons is 5 : 8.

Solution 2
Let the number of apples be $x$.
Since the ratio of the number of apples to the number of oranges is 1 : 4, then the number of oranges is $4x$.
Since the ratio of the number of oranges to the number of lemons is 5 : 2, then the number of lemons is $\frac{2}{5} \times 4x = \frac{8}{5}x$.
Since the number of apples is $x$ and the number of lemons is $\frac{8}{5}x$, then the ratio of the number of apples to the number of lemons is $1 : \frac{8}{5} = 5 : 8$.

23. In the addition of two 2-digit numbers, each blank space, including those in the answer, is to be filled with one of the digits 0, 1, 2, 3, 4, 5, 6, each used exactly once. The units digit of the sum is




(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

**Source:** 2006 Gauss Grade 7 #23

**Primary Topics:** Number Sense

**Secondary Topics:** Logic | Equations Solving | Digits

**Answer:** D

**Solution:**

We label the blank spaces to make them easier to refer to.

$$\begin{array}{cccc} & & \boxed{A} & \boxed{B} \\ + & & \boxed{C} & \boxed{D} \\ \hline & \boxed{E} & \boxed{F} & \boxed{G} \end{array}$$

Since we are adding two 2-digit numbers, then their sum cannot be 200 or greater, so $E$ must be a 1 if the sum is to have 3 digits.

Where can the digit 0 go?
Since no number can begin with a 0, then neither $A$ nor $C$ can be 0.
Since each digit is different, then neither $B$ and $D$ can be 0, otherwise both $D$ and $G$ or $B$ and $G$ would be the same.
Therefore, only $F$ or $G$ could be 0.

Since we are adding two 2-digit numbers and getting a number which is at least 100, then $A+C$ must be at least 9. (It could be 9 if there was a "carry" from the sum of the units digits.)
This tells us that $A$ and $C$ must be 3 and 6, 4 and 5, 4 and 6, or 5 and 6.

If $G$ was 0, then $B$ and $D$ would have to 4 and 6 in some order. But then the largest that $A$ and $C$ could be would be 3 and 5, which are not among the possibilities above.
Therefore, $G$ is not 0, so $F=0$.

$$\begin{array}{cccc} & & \boxed{A} & \boxed{B} \\ + & & \boxed{C} & \boxed{D} \\ \hline & \boxed{1} & \boxed{0} & \boxed{G} \end{array}$$

So the sum of $A$ and $C$ is either 9 or 10, so $A$ and $C$ are 3 and 6, 4 and 5, or 4 and 6.
In any of these cases, the remaining possibilities for $B$ and $D$ are too small to give a carry from the units column to the tens column.
So in fact, $A$ and $C$ must add to 10, so $A$ and $C$ are 4 and 6 in some order.
Let's try $A=4$ and $C=6$.

$$\begin{array}{cccc} & & \boxed{4} & \boxed{B} \\ + & & \boxed{6} & \boxed{D} \\ \hline & \boxed{1} & \boxed{0} & \boxed{G} \end{array}$$

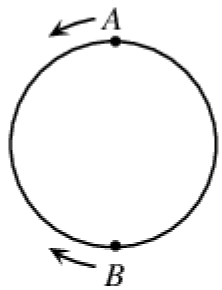
The remaining digits are 2, 3 and 5. To make the addition work, $B$ and $D$ must be 2 and 3 and $G$ must be 5. (We can check that either order for $B$ and $D$ works, and that switching the 4 and 6 will also work.)

So the units digit of the sum must be 5, as in the example

$$\begin{array}{cccc} & & \boxed{4} & \boxed{2} \\ + & & \boxed{6} & \boxed{3} \\ \hline & \boxed{1} & \boxed{0} & \boxed{5} \end{array}$$

(Note that we could have come up with this answer by trial and error instead of this logical procedure.)

24. On a circular track, Alphonse is at point $A$ and Beryl is diametrically opposite at point $B$. Alphonse runs counterclockwise and Beryl runs clockwise. They run at constant, but different, speeds. After running for a while they notice that when they pass each other it is always at the same three places on the track. What is the ratio of their speeds?




(A) $3:2$ (B) $3:1$ (C) $4:1$ (D) $2:1$ (E) $5:2$

**Source:** 2005 Gauss Grade 7 #24

**Primary Topics:** Algebra and Equations

**Secondary Topics:** Fractions/Ratios | Equations Solving

**Answer:** D

**Solution:**

Since Alphonse and Beryl always pass each other at the same three places on the track and since they each run at a constant speed, then the three places where they pass must be equally spaced on the track. In other words, the three places divide the track into three equal parts.
We are not told which runner is faster, so we can assume that Beryl is the faster runner.
Start at one place where Alphonse and Beryl meet. (Now that we know the relative positions of where they meet, we do not actually have to know where they started at the very beginning.)
To get to their next meeting place, Beryl runs farther than Alphonse (since she runs faster than he does), so Beryl must run $\frac{2}{3}$ of the track while Alphonse runs $\frac{1}{3}$ of the track in the opposite direction, since the meeting placed are spaced equally at $\frac{1}{3}$ intervals of the track.
Since Beryl runs twice as far in the same length of time, then the ratio of their speeds is $2:1$.

25. A school trip offered its participants three activities: hiking, canoeing and swimming. Attendance records show that of all participants
    - 10 students participated in all three activities,
    - $50\%$ participated in at least hiking and canoeing,
    - $60\%$ participated in at least hiking and swimming,
    - $k\%$ participated in at least canoeing and swimming, and
    - no students participated in fewer than two activities.

    If $k$ is a positive integer, what is the sum of all possible values of $k$?

    (A) 191 (B) 185 (C) 261 (D) 95 (E) 175

**Source:** 2023 Gauss Grade 7 #25

**Primary Topics:** Counting and Probability

**Secondary Topics:** Counting | Graphs | Logic

**Answer:** B

**Solution:**

We can represent the given information in a Venn diagram by first introducing some variables. Let $x$ be the number of students that participated in hiking and canoeing, but not swimming. Let $y$ be the number of students that participated in hiking and swimming, but not canoeing. Let $z$ be the number of students that participated in canoeing and swimming, but not hiking. Since 10 students participated in all three activities and no students participated in fewer than two activities, we complete the Venn diagram as shown.
Suppose that the total number of students participating in the school trip was $n$. Since 50% of all students participated in at least hiking and canoeing, then $\frac{50}{100}n$ or $\frac{n}{2}$ participated in at least hiking and canoeing. Since this number of students, $\frac{n}{2}$, is an integer, then $n$ must be divisible by 2.
Similarly, $\frac{60}{100}n$ or $\frac{3n}{5}$ students participated in at least hiking and swimming. Since this number of students, $\frac{3n}{5}$, is an integer, then $n$ must be divisible by 5 (since 3 and 5 have no factors in common).
This means that $n$ is divisible by both 2 and 5, and thus $n$ is divisible by 10. From the Venn diagram, we see that $x + 10 = \frac{n}{2}$, and $y + 10 = \frac{3n}{5}$. Since the total number of participants is $n$, we also get that $x + y + z + 10 = n$ or $z = n - 10 - x - y$. We may now use these equations,

$$x = \frac{n}{2} - 10,\ y = \frac{3n}{5} - 10, \text{ and } z = n - 10 - x - y$$

and the fact that $n$ is divisible by 10, to determine all possible values of $z$. We can then use the values of $z$ to determine all possible values of the positive integer $k$, where $k\%$ participated in at least canoeing and swimming.
Since $n$ is a positive integer that is divisible by 10, its smallest possible value is 10. However, substituting $n = 10$ into $x = \frac{n}{2} - 10$, we get $x = 5 - 10$ and so $x = -5$ which is not possible. (Recall that $x$ is the number of students that participated in hiking and canoeing, but not swimming, and so $x \geq 0$.) Next, we try $n = 20$. When $n = 20$, $x = 10 - 10$ and so $x = 0$.
When $n = 20$, $y = \frac{3 \times 20}{5} - 10$ or $y = 12 - 10$, and so $y = 2$.
Finally, when $n = 20$, $x = 0$, and $y = 2$, we get $z = 20 - 10 - 0 - 2 = 8$. When $z = 8$, the number of students who participated in at least canoeing and swimming is $8 + 10 = 18$ (since 10 students participated in all three), and so the percentage of students who participated in at least canoeing and swimming is $\frac{18}{20} \times 100\% = 90\%$, and so $k = 90$. In the table below, we continue in this way by using successively greater multiples of 10 for the value of $n$.
For values of $n$ that are greater than 100, we get that $z < 0$, which is not possible. Therefore, the sum of all such positive integers $k$ is $90 + 40 + 30 + 15 + 10 = 185$.